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AI Documentation: A path to accountability

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ABSTRACT

Artificial Intelligence (AI) promises huge potential for businesses but due to its black-box character has also substantial drawbacks. This is a particular challenge in regulated use cases, where software needs to be certified or validated before deployment. Traditional software documentation is not sufficient to provide the required evidence to auditors and AI-specific guidelines are not available yet. Thus, AI faces significant adoption barriers in regulated use cases, since accountability of AI cannot be ensured to a sufficient extent. This interview study aims to determine the current state of documenting AI in regulated use cases. We found that the risk level of AI use cases has an impact on the AI adoption and the scope of AI documentation. Further, we discuss how AI is currently documented and which challenges practitioners face when documenting AI.

1. Introduction

Artificial Intelligence (AI) promises enormous potential for multiple industries (McKinsey Global Institute, 2018). What makes AI, and particularly AI's subfield Machine Learning (ML), so powerful is that AI models derive their structure from the data and need significantly less input from developers than traditional statistical models (Huang et al., 2004). This characteristic in combination with significant increases in the amount of available data and the reduction in the cost of computing power enables AI to uncover and use previously unidentifiable patterns in datasets (Halevy et al., 2009; Sun et al., 2017). This unique ability of AI enables companies to improve existing processes and models (Thalmann et al., 2018), and to create new business models (Fruhwirth et al., 2021). In medicine, for instance, AI applications can detect and diagnose cancer and heart diseases (Esteva et al., 2021). In consumer lending, highly accurate AI models can help lenders reduce defaults (Kvamme et al., 2018; Thalmann et al., 2018) or in production to predict machince failures (Gashi et al., 2021).

However, AI's ability to learn from data without requiring instructions from developers also makes explaining the reasoning behind some AI models to customers, auditors and regulators challenging. This characteristic of AI is particularly challenging in regulated industries, where transparency and accountability are required (Kloker et al., 2022; Königstorfer and Thalmann, 2021; Kvamme et al., 2018). The predictive power of convolutional neural networks, for example, is significantly higher than the predictive power of traditional statistical models, but the inability to explain convolutional neural networks makes it very challenging to implement convolutional neural networks in regulated use cases such as consumer lending (Kvamme et al., 2018). As a result, tools and methods for verifying that AI applications work correctly and behave in line with all legal or compliance requirements are needed.

In the IT industry, IT audits, or software validation procedures focus on ensuring the correct functioning of information systems (IS) and their compliance with regulations and industry standards. For that, IT auditors typically rely on suitable documentation of the IS (Bertl et al., 2019; IEEE, 2008). In the software industry, software documentation has proven to be essential for investigating and validating the proper functioning of the software (Bertl et al., 2019; IEEE, 2008). To ensure that developers can provide a suitable documentation to auditors, standardized documentation guidelines and audit procedures for traditional software systems exist (Hayhoe, 2012; IEEE, 2008; ISO, 2018). Thus, AI documentation seems to be a promising starting point for auditing or validating AI. However, due to significant differences between traditional software and AI, standard documentation guidelines and IT audit processes for conventional software systems are less meaningful for auditing AI (Holstein et al., 2019; Königstorfer and Thalmann, 2020, 2021).

However, implementing suitable AI documentation has received limited attention in the literature. Recent research has shown that current documentation guidelines for AI do not provide sufficient guidance for practitioners and that new guidelines are needed (Königstorfer and Thalmann, 2021). As a consequence, it is very challenging to audit or validate AI which represents an adoption barrier of AI in general and in regulated use cases in particular (Kvamme et al., 2018; Polzer et al.,

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2022). Thus it seems relevant to investigate the current state of the art of AI documentation and its impact on the adoption of AI. This leads to the following research question:

How does the need for documenting AI affect the adoption of AI?

The paper is structured in the following way. Section 2 discusses the existing literature. Section 3 presents the methodology for the paper. Section 4 presents the results of the study. Section 5 discusses the results and proposes avenues for further research. Section 6 concludes the paper.

2. Background

In recent years, powerful technological innovations have changed the IS landscape. In the last two decades, combining more available data and more affordable computing power has enabled AI to solve more complex tasks (Halevy et al., 2009; Sun et al., 2017). However, AI applications are subject to strict regulations in sensitive use cases, e.g. EU AI proposal for harmonized rules on AI and the US' proposed standard for identifying and managing bias in AI (European Parliament, 2021; National Institute of Standards and Technology - U.S. Department of Commerce, 2022). Due to the black box character, however, it is currently very challenging to audit or validate AI systems which represents a significant adoption barrier in sensitive and regulated use cases such as pharmaceutical production (Polzer et al., 2022) or commercial banking (Königstorfer and Thalmann, 2020).

In software engineering, the evaluation and verification of the proper functioning of the software and the conformance of software products to regulations, guidelines, and standards are verified during IT audits (IEEE, 2008). During IT audits, auditors examine all software components that can impact the proper functioning of the software, such as the service description, IT landscape description, requirements description, control description, and transaction logs (Bachlechner et al., 2014). During the examination step of the audit, IT documentation plays an essential role (IEEE, 2008). The documentation allows auditors to assess, for instance, how different system components and user policies have been implemented, how disaster recovery plans have been designed, and how users interact with the software system (Chudasama et al., 2019). Research has shown that suitable documentation of AI also forms the basis for internal and external audits of AI applications (Ellul et al., 2021; Raji et al., 2020), making it an auspicious starting point for auditing AI.

In software engineering, documentation is the *creation and keeping current of an artifact holding information about architectural decisions for stakeholders in an accessible, nonambiguous form* (Clements et al., 2011). This documentation can serve several purposes and be targeted at several audiences, including software engineers and IT auditors (Clements et al., 2011). Software engineers, for instance, depend on the system's source code combined with the technical documentation to create and maintain software (Garousi et al., 2015). The technical documentation is particularly important since it presents information on processing information, including internal controls and error correction mechanisms. To avoid the time invested in documentation becoming a waste of time, specific quality criteria need to be fulfilled (Parnas, 2011). Zhi et al. (2015) found that writing complete, accurate, and correct software documentation is not enough and identified a set of 15 quality criteria for software documentation. On the other hand, IT auditors also need to analyze the user documentation (Bertl et al., 2019). However, since the user documentation provides information on the overall functionality of the software from the perspective of the user (i.e., input fields, general description of the software) and does not discuss the implementational details of the software system, this type of documentation is not the primary focus of this paper.

To ensure that these requirements are met and that the resulting documentations are comparable, documentation guidelines were developed by organizations such as IEEE and ISO (ISO, 2018). Even though following such documentation standards is not always legally required, research argues that many challenges can be avoided by following a standard documentation process (Satish and Anand, 2016). However, research has shown that current approaches for documenting software are less meaningful for the documentation of AI and that more suitable guidelines for documenting AI are needed (Königstorfer and Thalmann, 2020, 2021).

The reason behind the reduced meaning is that AI, and particularly AI's powerful subfield ML are very different from traditional software. First, unlike conventional software, programmers do not tell ML models how to make their decisions but present an ML algorithm with data and tell the ML algorithm what it needs to know (Rodovold, 1999). As a result of ML's unique ability to derive its structure from the data, the source code of ML algorithms is less meaningful for the AI system than for traditional software. Even though this characteristic can have several benefits in terms of predictive power (Butaru et al., 2016; Khandani et al., 2010; Martens et al., 2016) and usability in practical use cases, it also presents a challenge when it comes to explaining and documenting the decision-making process of the AI (Gashi, Vuković et al., 2022). For instance, the inability to explain some AI models' decisions is known as the black-box character of some AI models (Vuković and Thalmann, 2022). This black box character is a barrier to implementing some AI models in practice (Kloker et al., 2022; Königstorfer and Thalmann, 2021; Kvamme et al., 2018). The limited meaning of the source code also poses a challenge for traditional software documentation guidelines. Existing research has shown that the source code is one of the preferred types of documentation (Garousi et al., 2015). As a result, challenges regarding the collection of suitable audit evidence in the Big Data and AI environment arise (Appelbaum et al., 2017). New, AI-specific documentation guidelines for AI are needed.

Second, the development of AI models highly depends on the data used to train the AI. This can lead to unintended and illegal biases in the decision-making of the AI without the AI developer realizing that the biases are there (Barocas and Selbst, 2016). Additionally, the AI can make errors if presented with data different from the data it was trained on (Ellul et al., 2021). The training data does not receive much attention in software documentation guidelines for audit purposes (Bertl et al., 2019).

Third, human influence on data and AI is often not reported adequately (Diakopoulos, 2016; Miceli et al., 2020; Miceli et al., 2021). This does not only mean that human biases can prevail in algorithmic decision-making but also that AI could be used in an unintended manner (Arnold et al., 2019), which can lead to incorrect predictions, automation biases and potentially harmful consequences (Kloker et al., 2022).

Fourth, researchers point to recent changes in the structure of IS (i.e., cloud- and distributed computing systems), changes in the development of AI (i.e., frequent, near-continuous re-training), and the near-constant inflow of new data can cause implementations of ML to violate several theoretical assumptions underlying ML algorithms (Parker, 2012). Hence, the use of these new technologies can have a significant impact on the performance and the business impact of the AI application. Documentation guidelines need to be adapted to incorporate information on these aspects.

New documentation guidelines have already been proposed to communicate the inner workings of AI applications understandably and enable effective AI audits. For instance, guidelines specifically focusing on the model development (Mitchell et al., 2019; Rodovold, 1999) and the documentation of the training data exist (Gebru et al., 2018; Miceli et al., 2021). However, significant practical challenges for deploying AI in practice, such as integrating AI into business processes, remain unaddressed in these proposals. Further, researchers also proposed extensions to existing standards for reporting the results of clinical trials that investigate the use of AI in medicine (Liu et al., 2020; Rivera et al., 2020). The extensions to these guidelines argue for the inclusion of the characteristics of the AI model itself as well as the integration of the AI into the trial setting, the interaction between the AI and humans, and the actions that are taken based on the AI's recommendation (Liu et al., 2020; Rivera et al., 2020). However, the authors point out that medical professionals involved in clinical trials do not yet follow the extensions to the guidelines (Rivera et al., 2020). Also, the authors do not suggest how the frequent re-training of AI needs to be documented (Liu et al., 2020; Rivera et al., 2020), even though this characteristic is already being used by several practitioners (Barque et al., 2018). This means that it remains unclear whether the guidelines are known and in practice or whether other methods for documenting AI are being used.

3. Method and procedure

To investigate the current way of documenting AI in practice, an exploratory study with semi-structured interviews was conducted. Three groups of professionals were selected: First, IT professionals in banks and regulators of banks on multiple levels were targeted. Use cases from banks served as examples for regulated use cases. The fact that the documentation of AI was a barrier to the implementation of AI in banks was already identified in the literature review (Königstorfer and Thalmann, 2020). Second, AI engineers from companies that specialized in AI were contacted. The expertise in the technological aspects of AI and their broad knowledge of AI use cases was considered valuable for the study. Third, consultants and regulators in the banking industry were selected for the study as their unique insights into the regulatory framework complement our focus on regulated use cases. Two rounds of interviews were conducted. The first round aimed at collecting primary insights into the focus of the study, while the second round focused on validating the findings of the first round.

3.1. *First round of interviews*

For the first round, 16 interviewees were recruited based on a convenient sampling and interviewed via Skype for Business between March and June 2020. Four interviews were conducted in English, and the remaining 12 were in German. An interview guideline was created based on a structured literature review on the existing use cases of AI (Königstorfer and Thalmann, 2020). The interview guideline was designed to elicit the current state of using and documenting AI as well as the most essential and challenging requirements for the documentation of AI. Further refinements to the interview guide were made after a pre-test interview. The interview guide that was used during the interviews can be found in Appendix A.

Table 1 shows details about our interviews and the industry sector in which they work. In the interviews, the experts talked about sensitive use cases such as fraud detection in the financial industry or self-driving cars. Five interview partners had between three and ten years of experience in a relevant role, while the remaining eleven interview partners had more than ten years of experience in an appropriate position. Interviewees 1 and 4 were working at the same software company, and interviewees 11 and 16 were working in different offices of the same international company at the time of the interview. The interviews were recorded, transcribed, and cleaned. The interviews lasted 34 minutes on average. In Table 1, basic information about the interviewed experts can be found. Except for two interviewees, all interview partners worked in senior positions, such as CEO, CTO, or Head of AI. Most interviewees had been exposed to AI during their education as well as in their workplace. Hence, the interviewees knew both the technical side of AI and the impact of AI on the business side of the company well.

Based on the transcripts from the first round of interviews, a qualitative content analysis using an informed inductive coding approach was conducted (Patton, 2014). MAXQDA was used for the coding and initial codes based on categories from a previously conducted structured literature review (Königstorfer and Thalmann, 2020). For instance, the structured literature review revealed that the data used for training the AI was essential. Therefore, the term "description of data" became a code. During the coding, new codes emerged. For example, the

Table 1
Interview partners in the first round of interviews

Expert ID	Industry	Country of employment	Professional experience	Language in which the interview was conducted
1	Software	Austria	> 10 years	German
2	Banking and Finance	Austria	> 10 years	German
3	Banking and Finance	Germany	> 10 years	German
4	Software	Austria	> 10 years	German
5	Software	Austria	3 – 10 years	German
6	Software	Austria	> 10 years	German
7	Consultancy	Finland	> 10 years	English
8	Consultancy	Austria	3 – 10 years	German
9	Banking and Finance	Germany	> 10 years	German
10	Banking and Finance / Regulator	Germany	> 10 years	German
11	Consultancy	Germany	3 – 10 years	German
12	Banking and Finance	Netherlands	> 10 years	English
13	Banking and Finance	Switzerland	> 10 years	German
14	Consultancy	Netherlands	> 10 years	English
15	Banking and Finance	Austria	> 10 years	German
16	Consultancy	Austria	3 – 10 years	German

interviewees stated that the data preparation and cleaning were just as necessary as the data itself. Hence, "data cleaning and preparation" became a code as well. After the coding, the codes "description of data" and "data cleaning and preparation" were categorized as "description of training data and data preparation." After that, the quotes used in this paper were translated, and both authors checked the translations. The requirements for AI documentation were extracted in the same manner.

During the analysis, several insights were generated. The primary findings were four different groups of AI documentation beneficiaries. However, questions regarding the suitability of the grouping as well as the applicability of these requirements in regulated use cases remained. To answer these questions, the second round of interviews was conducted.

3.2. *The second round of interviews*

In the second round, four additional evaluation interviews were conducted between April and June 2021. Two interviews were conducted in English, and the remaining two were in German. All but one of the interviews in the second round were recorded and transcribed. In one of the interviews, the recording failed to start due to a technical error, and a memory protocol was produced immediately after the end of the interview.

The preliminary findings from the first round of interviews are presented in the form of a PowerPoint presentation. The interviewees were then asked to give feedback and to critically reflect on the findings, which resulted in detailed discussions. The code set from round one serves as a starting point for coding the second round of interviews.

Based on the feedback from the evaluation interviews, a few minor changes were made. For instance, the interviewees recommended renaming the group "Risk Averse non-adopters to "Cautious non-adopters".

4. Findings

Given the scope of the study, we initially targeted users of AI in regulated use cases. However, the coding revealed apparent differences between the use cases that the interviewees described, particularly along two dimensions: (1) the level of risk associated with the use case; and (2) the adoption of AI. The level of risk is defined by a set of regulations from bodies of the European Union. Specifically, the General Data Protection Regulation (GDPR) (European Commission, 2017; European Parliament, 2016) and the recent proposal for the regulation of AI in practice (European Parliament, 2021) provide definitions for "high risk" use cases of processing techniques.

The adoption of AI is a categorical feature that indicates whether an interviewee talks about an existing application of AI in an organization after being asked to present one. Interviewees who state that they do not use AI in their organization are considered non-users of AI, while interviewees who use AI in their organization are regarded as users of AI.

Consultants, software developers, and regulators do not talk about applications of AI in their organizations but about use cases of AI in other companies. To ensure that the different approaches to documenting AI can be presented appropriately, the interviewees were grouped based on how they described the potential use cases of AI. Every interviewee can be assigned to exactly one group. These two dimensions are the result of the coding. Table 2 below displays the relationship between the two dimensions and the resulting types.

The distinction between the different groups is essential for the findings of this paper. Due to the differences in the risk level, AI adopters document their AI applications differently.

The first group of interviewees, the *cautious non-adopters*, are interested in adopting AI and have already experimented with the technology. However, these experiments did not result in products or services that were adopted in practice. The primary reason reported was that interviewees do not know how to fulfill all compliance requirements related to AI. The high criticality level, in combination with the lack of suitable documentation guidelines for AI, makes it challenging for the interviewees to apply AI in such use cases. Due to the lack of suitable documentation guidelines and auditing procedures, the interviewees are considering ways to circumvent the existing compliance requirements. Two of our interviewees from the banking and finance industry fall into this group, and one of the consultants talks about clients in this group.

> "*Maybe I am working in a medical company, and because it is in the interests of my patient and* ***because not everything has been answered yet,*** *I am set in such a way that I say: "There is currently no sensible implementation for me to ensure that that the AI is easy to implement."* " (Evaluation Interview 4)

The second group of interviewees, the *relaxed adopters*, have adopted AI in their organization or described use cases of AI in practice. The risk level and the probability of getting audited in the discussed use case are comparatively low. Additionally, the interview partners point to the trade-off between the creation of lengthy, detailed documentation on the one hand and the protection of company-internal knowledge and the innovativeness of the company on the other hand. For this reason, interviewees of this group have decided to center their AI documentation on the education of their team. In this type of use case, the hands-on approach to documenting AI is characterized by its focus on the development process. Four interviewees from the banking and finance industry and two AI specialists in software companies fall into this group, and one of the consultants talks about clients in this group.

The third group of interviewees, the *prepared adopters*, use AI in use cases with a high-risk level. Just like the relaxed adopters, the prepared adopters also use AI in their organization or describe use cases of AI. The interview partners not only attempt to keep the users of the AI safe, but they also feel responsible for the ethical implications of their AI applications. To ensure that the documentation can meet all these requirements, the interviewees have developed an AI-specific documentation approach that not only focuses on the development of the AI itself but also takes the context of operation into account. One interviewee from the banking and finance industry and two AI specialists in software companies fall into this group. Three of the consultants talk about clients in this group.

The fourth group of people, the *unexcited non-adopters*, is not interested in adopting AI and has not thought about the difference between AI and traditional software systems or related compliance concerns. As our goal was to investigate the requirements for AI documentation and how AI is currently documented in regulated use cases, we could not expect insights from this group. Thus, this group was not targeted for this study, and no interviewee could be assigned to this group. We invite researchers to analyze and characterize this group in a more detailed manner.

The classification (see Table 2) and its dimensions were discussed with experts during the evaluation interviews. The interviewees supported our proposal and especially the case-wise distinction as one expert highlights:

> "***That makes perfect sense to me*** *because documentation depends on your case and each case needs different documentation.*" (Evaluation Interview 3)

4.1. Cautious non-adopters: AI is currently not used due to high criticality levels and high perceived compliance requirements

The first group of interviewees, the cautious non-adopters, state that they primarily work in use cases with a high level of risk and regulation. They perceive the compliance requirements in the type of use case they describe as high. This type of potential use case of AI is also subject to a high degree of regulatory oversight. Organizations not only need to ensure that their AI documentations are up to date for potential audits, but they also often need to get their AI solutions certified before using their AI-based system in this type of use case. Expert 3, a compliance officer at a bank, describes his situation in the following way:

> „*We are a … bank, and the ECB will quickly say: "We will* ***not allow you to implement this AI-based system."*** *We can't keep up with that, and I think that's why there are not so many things in the field that you are working in*" (Expert 3)

This shows that the high compliance requirements and the high degree of oversight represent a barrier to the implementation of AI for this group of interviewees.

One primary reason for not being able to meet the compliance requirement is the lack of suitable documentation guidelines. Existing documentation approaches for traditional software and statistical models are less meaningful for the documentation of AI. Less meaningful means that core aspects needed to reproduce functionalities of the AI-based system in an IT audit are missing. Due to the significant differences between the traditional statistical models and AI, existing documentation methods do not address the specific challenges of AI adequately, as Expert 8 illustrates:

> "*If [you] take another step towards neural networks or deep learning, then you would have to document the processing of the data inside of these neural networks accordingly. And when it comes to the complexity [of documenting the processing of the data inside the neural network], that is a completely different challenge, where* ***I am currently doubting whether it can even be described sufficiently well in the current form of the documentation.*** *[…]In my opinion, this is also one of the big aspects that probably speaks against the use of such models*" (Expert 8)

In the interviews, bankers, consultants, and employees of software companies revealed that they struggled with describing AI solutions in

Table 2
Types of AI documentation users

		Risk level Low risk	High risk
Adoption of AI	No	Unexcited non-adopters	Cautious non-adopters
	Yes	Relaxed adopters	Prepared adopters

such a way that they can be audited, particularly with more powerful neural networks and deep learning algorithms. Additionally, the interviewees point to a lack of suitable regulatory guidelines for the documentation of AI and a mismatch between the regulatory requirements and the characteristics of AI. Interviewees miss explicit guidelines on how to create and how to document AI to comply with compliance requirements.

Even though organizations can implement AI-based services and have promising use cases, the fact that they cannot describe AI sufficiently for an audit prevents their adoption. Expert 3 explains this:

> "*So there we are in these innovation labs, which everyone is doing at the moment, and has been doing for a long time. And there we talk about all the great things that we can do with AI. And those things work well. [...] However, we already noticed that the* ***problem is getting them up and running in the bank because they often do not meet the compliance requirements*** *at all. And that's why we fail most of the time*" (Expert 3)

This shows that the lack of meaningful documentation guidelines is a barrier to the adoption of AI in practice.

This adoption barrier puts banks at a disadvantage compared to less regulated FinTechs, which do not have to comply with such strict compliance requirements. However, interviewees do see the value of AI and think about methods to mitigate the situation. One expert outlined a mitigation strategy:

> "*Before* ***we are not using the new [AI-based] services, we found start-ups and purchase their services.*** *The fin-techs have much lower regulatory requirements. We as banks also would like to do this, but we are not allowed to do so*" (Expert 3)

This quote clearly shows the wish to introduce AI and that companies are searching for ways to circumvent the perceived regulatory pressure.

4.2. Relaxed adopters: AI is used in use cases with low criticality levels, and AI is documented hands-on

In contrast to the previous group of interviewees, the relaxed adopters apply AI in use cases with low levels of risk. The interviewees in this group argue that the compliance requirements in their use case are not as strict as in other types of use cases, the probability of getting audited by a regulator is lower, and there is little or no pressure to comply with internal ethical guidelines. Additionally, none of the AI applications in this type of use case need to be pre-approved by a regulatory agency. Expert 15 contrasts the situation that he is in with other types of use cases in banks in the following way:

> "***[A documentation is] currently not required****, because it is a purely internal managerial model. The subject will be a completely different one if we try to develop a model that is also approved by a ... the European Central Bank*" (Expert 15)

As a result of the risk and the low criticality, organizations have much more freedom to use AI in this type of use case. Thus, AI is being applied in use cases with no or only limited connection to customer data (i.e., prediction of ATM failures; portfolio management; market-making) or with low perceived oversight (i.e., marketing, fraud detection). While some interviewees simply replace traditional statistical models with more accurate AI models, others decide to integrate AI into an existing, static business process or create new, fully automated business models. Interviewees also state that they are at different levels in their AI development process. While some interviewees are already using AI-based services, others are still working on creating viable prototypes.

Since compliance requirements seem less relevant to the interviewees, developers use a hands-on approach to documenting AI. The interviewees use an annotated version of their code as documentation for their AI. Expert 9, for instance, describes his method for documenting AI in the following way:

> "***We started using the code together with markup directly as documentation.*** *This means that the markup file contains the background as to why the developer has decided to take certain steps. It really starts at the very beginning, with the sample. […] and then the SQL code that pulls the data from the database is embedded in the document. Then comes the next step, where you say: Okay, we'll use [this code] to train features of different nature. These are then briefly described in words. [...] This is how it is described, and then the R code follows, which includes the data that has emerged from the SQL query and the code that converts the raw data into these features. And that is how it goes on. […] The model itself is actually the most unspectacular thing. [...] The documentation would describe exactly what algorithms we tried, including the ones that did not lead to the desired result. [...] And then we would describe the models that led to success. And yes, that in Markdown, and then the R code that was actually used to train the model would follow. With the training parameters and so on and so on. And then finally a few comments about what the developer saw and where the final product lies*" (Expert 9)

The intense focus of the documentation on the development process and the performance tests of the AI model makes this documentation approach very technical and hands-on. Interviewees from banks, software companies, and consultancies noted that the development process description is an essential requirement for suitable documentation of AI. The interview partners argue that the primary beneficiaries of this AI documentation are AI developers and that the documentation, therefore, needs to be understandable for AI developers. Descriptions of the IT landscape, the controls of the AI, the application context, and the transaction logs are not part of this approach to documenting AI. This, however, would require additional tools and processes when the AI is audited.

The approach for documenting AI enables the development team to remain innovative while enabling organizations to maintain their IS in the long run. Expert 9, for instance, argues that he wants to ensure that new employees can learn how the AI works quickly.

> „***This documentation*** *that I have just described, is perfect to ensure that someone can understand how the model was developed. For example, if a developer gets sick for a long time or leaves the company for some reason and someone else has to familiarize themselves with the topic, then they can do that.* ***That works very well***" (Expert 9)

Creating understandable AI documentation is a requirement for the documentation. It is important to note that the interview partners document their AI applications for their employees and ensure that their development team stays innovative. At the same time, interviewees point to the trade-off they face, namely the trade-off between protecting their business-critical knowledge and working in a resource-efficient manner and ensuring that the AI is documented sufficiently.

Several interviewees also point to the lack of suitable AI documentation and audit guidelines from regulators and lawmakers. This causes significant uncertainties for developers and internal auditors and causes practitioners to delay the documentation process and to write non-standardized and potentially unsuitable documentation for AI.

> „*The challenge will also be that the* ***current units that check the models,*** *such as the validation- and the internal auditing department****, do not yet know how to approach this activity.*** *That means not only me as a developer, but also a validation- and an auditing department first have to consider the methods and approaches with which they want to test a developed model. That has to be done first. And that, is the reason why* ***I am currently waiting with the documentation***" (Expert 15)

Even though AI developers want to document AI in a way that does not only comply with legal requirements but also provides all relevant information for audits, creating suitable documentation is not always possible. The primary reason behind this inability to create suitable documentation is the uncertainty and unclarity on how AI is audited and on what needs to be documented to pass an audit. More work needs to be done on AI documentation guidelines and AI audit standards.

4.3. Prepared adopters: AI is used and documented in an AI-specific manner

The third group of interviewees, the prepared adopters, apply AI in use cases with a high level of risk. The interviewees perceive the compliance requirements as high. While interviewees from banks and consultancies primarily point to the regulatory requirements they need to comply with, interviewees from software companies point to their sense of responsibility and argued that complying with a set of principles is equally important.

> „***We have defined principles as to how we believe AI models should work, and the documentation is also based on these principles.*** *The documentation is, therefore, our tool to implement the ethical principles in the best possible and most credible way*“ (Expert 4)

In both cases, interviewees consider the risk as very high and make meeting the necessary compliance requirements a top priority.

The AI applications of the interviewees are subject to such strict compliance requirements because of the processing of personal data (i. e., in recruiting processes), the attention that regulators pay to the use cases (i.e., internal audit of customer accounts), and the strong negative impact of a malfunction of the AI could have on people's lives (i.e., self-driving car). Most interviewees use AI to make incremental changes to an existing business process or products (i.e., extracting relevant information from customers' E-Mails). Some interviewees also describe use cases in which AI is used to make significant changes to an existing product or business process (i.e., making cars drive by themselves or using a chatbot in recruiting). Similar to the relaxed adopters, this group of interviewees uses AI internally, as Expert 12 explains:

> "*We* ***use machine learning/AI for audit tasks****. So finding anomalies in data sets for internal audits and anomaly detection for transactional risk profile of our customers*" (Expert 12)

This shows that even though there are high compliance requirements, this group of interviewees is willing to implement AI.

The prepared adopters see the documentation as a method for ensuring compliance with ethical and regulatory requirements. For this reason, they create AI-specific documentation. In addition to the aspects mentioned in the previous subchapter (data, feature engineering, feature descriptions, the algorithm that was used to train the model, unsuccessful algorithms, training parameters, performance tests, location where the final model is stored, and comments from the developer), the AI documentation of this group (1) contains information on the business, and IT environment in which the AI operates, (2) automates the documentation of several technical aspects related to the development and the IT environment of the AI, (3) documents the explanations of the AI and (4) documents all aspects in a higher degree of detail than the documentation in the other types of use case. Expert 12 explains:

> "*So how do I document?* ***That's the way the platform documents all the actions an engineer took during the creation of a model.*** *So the platform loggs every action from a… from my engineer. The platform itself delivers an explanation for every decision it makes, even though it's supervised or unsupervised. And then you come to the documentation. Documentation for me is: "Have you documented the question that you wanted to address with this AI/Machine learning approach? Have you documented the data? So, where's your data coming from? And did you and what testing that you did you do to ensure that what it's what the algorithm does is contributing to the business question. That's where I am now.*" (Expert 12)

Interviewees attempt to meet the regulatory and ethical standards using this documentation. This approach to documenting AI enables the interviewees to meet various regulatory and ethical requirements. Documenting the business- and the IT environment in which the AI operates enables users and auditors to understand the potential downstream effects of the AI, which is essential in the context of an audit. The interviewees highlighted that information about the business environment in which the AI operates and especially how AI outcomes are embedded into business processes is an essential requirement of AI documentation. These interviewees also highlighted that such information is even more important from their perspective than the technical details of the AI.

Since parts of the documentation process have been automated, developers have little to no opportunities to hide unwanted actions. Even though the interviewees point out that not all parts of the documentation can be automated, they did argue that automation saves a significant amount of time. One aspect that the interviewees were particularly proud of was the fact that the biases in the data and the technical decisions around the architecture of the model were documented, as one interviewee points out:

> „*In principle,* ***AI applications should be documented to the extent that their results are reproducible.*** *In this regard, one must, among other things, keep track of which training data was used for training the AI model. Concerning the training data, one should also record as to whether there were imbalances in the training data in the categories to be trained or how these imbalances were corrected. […] In addition to the information regarding the data, technical decisions such as the architecture of the model, as well as the algorithms and hyperparameter settings used should be documented in a comprehensible manner*“ (Expert 4)

Documenting these additional technical details of the development as well as the performance of the AI helps ensure that the AI behaves ethically. For instance, the documentation can be used to prove to an auditor that potentially discriminatory patterns in the data and the decision-making process of the AI have been taken care of. Additionally, interviewees argue for making the predictions of AI as transparent as possible. As a result, biases that could distort the decision-making process of the AI could be detected earlier. Interviewees pointed out that documenting major design decisions is an essential requirement for the documentation of AI.

However, the lack of clear documentation guidelines and audit procedures for AI also represents a challenge for this group. Primarily, the interviewees highlight that despite their tremendous efforts, it is not clear whether the AI documentation complies with regulations and FAT AI requirements. Expert 12 argues in the following way:

> "***We still do not know whether we have the right documentation if regulators come in*** *and do audits. The* ***guidelines that are out there are very vague.*** *And sometimes the documentation questions go beyond what you could know. […] What do I document? Do I document the process or the outcome? And if the outcome is ever-changing, do I need to document a changing outcome?*" (Expert 12)

Interviewees point to the existing privacy laws that prohibit sharing personal data with people outside of an organization. This also means there is a lack of clarity around what the documentation of the development process needs to look like. Managing the trade-off between sharing all relevant information about the development of AI and privacy laws is still an open challenge. Practitioners would benefit from suitable documentation guidelines. Interviewees also point to ongoing conversations with lawmakers and regulatory agencies on the topic of AI documentation.

This shows that this group of interviewees chooses to document AI in such a way that the IT and development process, the performance, the decision-making process as well as the business context of the AI become more transparent. To keep efforts within certain limits, they push automated documentation tools to keep track of the essential AI development decisions and changes.

5. Discussion

In our study, we found three different perspectives on how to document AI. Thereby, we found that the perspective mainly depends on (1) the risk level of the use case; and (2) the level of AI adoption. The risk level is based on EU-wide data- and AI regulations (European Commission, 2017; European Parliament, 2016, 2021). However, this classification is also consistent with other publications and certification procedures. Winter *et al.* (2021), for instance, argue that certification procedures for AI applications need to differ depending on the amount of harm a failure of the AI can do to humans. The authors define four criticality levels. The first and the second criticality level defined by Winter *et al.* (2021) could correspond to the "low" level of criticality defined in the regulations mentioned above, while criticality levels three and four defined by Winter *et al.* (2021) could correspond to the "high" level of risk in the regulations. In reaction to the new EU AI regulation and their definition of use cases, we expect more research in this direction, and we see our work as a valuable starting point. Our results are summarized in Table 3.

The first group of interviewees, the cautious non-adopters, do not adopt AI in their use cases. The interviewees talk about use cases with high levels of risk. The core argument is that they cannot get AI audited and thus do not fulfill the compliance requirements. Documentation is an essential issue for the non-adoption of AI in this regard, particularly the missing documentation and auditing guidelines. The interviewees also talk about the benefits AI could deliver and present interesting workarounds to the challenges related to applying AI in practice. Interestingly, these workarounds are also in line with existing research. The bank BBVA argues that it did not find a way to integrate its Data Science and AI department in the bank and decided to outsource its Data Science and AI efforts to an independent subsidiary (Alfaro et al., 2019). Thus, we provide a more detailed perspective on scholarly work, arguing that limited explainability, i.e., Kvamme *et al.* (2018) and Gashi et al. (2022) or missing documentation guidelines, i.e., Königstorfer and Thalmann (2020), is a barrier to the adoption of AI. Particularly the initiatives to bypass strict regulations (i.e., by founding start-ups) underline the potential of AI and the pressure use case owners perceive.

The second group of interviewees, the relaxed adopters, apply AI in use cases with a low level of risk. Interviewees document AI in a hands-on manner, mainly to satisfy technical and team coordination needs. As these interviewees are not confronted with strict compliance requirements in their use cases and do not expect to be audited, they do not feel the pain of missing documentation guidelines. Instead, they perceive documentation as a burden and only see themselves as consumers of documentation. Being asked about the suitability of their hands-on approach to documentation, the interviewees are aware of the limitations, and they argue for the protection of intellectual property. Thus, even if the interviewees work in regulated industries, their use cases are not subject to high regulation. This is in line with research focusing on the adoption of AI, showing that AI is already applied in some areas of banks, such as marketing (Martens et al., 2016), and the management of internal operations (Lázaro et al., 2018). Our findings

Table 3
Summary of results

	Unexited non-adopters	Cautious non-adopters	Relaxed adopters	Prepared adopters
Documentation approach	No documentation of AI was written due to the non-adoption of AI.		The code, in combination with a description, is the documentation.	Documents containing the documentation exist. The documentation is partially automated.
Type of use case	Use cases with low risk.	Use cases with high risk. The application of AI inside the existing organization is perceived as impossible.	Use cases with low risk. Example: Predictive Maintenance	Use cases with high risk. Example: Internal Audits
Content	No documentation of AI was written due to the non-adoption of AI.		Code that pulls the data from the database Features and description of features in words Feature descriptions + code used to convert the raw data into features Performance tests and algorithms that were tested (models that failed as well as the one that was chosen) For the model that was chosen: the training parameters Comments about any interesting insights that were generated during development Where the final model is used	In addition to the aspects documented by the relaxed adopters, the following aspects are documented: Business and IT environment in which the AI operates (incl. business questions and integration into the business process) Logs of every action conducted by the engineer during the development process An explanation for every decision of the AI and the development platforms Additional details on (1) Data collection process and (2) Treatment of imbalances and other errors in the dataset
Users of the documentation		Auditors and regulators	Development team	Auditors and regulators
Benefits of the documentation		Documentation could help with getting AI applications through regulatory audits	Helpful in training new employees Model development is easy to understand	The documentation contains a lot of detailed information
Challenges related to the documentation		AI documentation guidelines are unknown or unclear	The trade-off between protecting business-critical knowledge and working in an efficient manner Ensuring that the AI is documented sufficiently Departments that check the models (validation and internal audit department) do not yet know how to audit and validate AI	Existing documentation standards for AI are too vague. It is unclear whether the existing documentation guidelines are sufficient for regulators Remaining unclarities around the documentation of some technical aspects, such as the documentation of frequent re-training of the AI The trade-off between sharing data documentation and privacy laws was not addressed.
Primary basis/ Requirement for the documentation			Usefulness and understandability	Ethical guidelines Regulatory requirements

thus call for a more detailed analysis of AI adoption and focus on use case characteristics, such as criticality instead of industry sectors.

The third group of interviewees, the prepared adopters, use and document, AI in use cases with high levels of risk. They developed documentation approaches taking the specifics of AI and the integration of the AI into the business and the IT environment into account. This is in line with recent research, showing that the specifics of AI deployment and related issues such as the integration into business processes and AI management concepts are significant concerns for practitioners and researchers (Benbya et al., 2021). However, even if they developed specific documentation guidelines and invested a lot of effort, they do not know if their documentation fulfills the requirements of audits. As such, an audit is mandatory from a legal perspective (leading to non-adoption) and is desirable for ethical reasons. Thus, we could show that even forerunners in the field of AI and AI documentation face real challenges when it comes to auditing and that audit guidelines are needed. Further, we revealed that practitioners are aware of the difficulties of documenting AI and spend tremendous efforts developing suitable approaches. However, industry standards, guidelines, and recommendations are missing, even as the importance of ethical and legal guidelines in the context of AI has grown in recent years (Jobin et al., 2019; Smit et al., 2020). Just like interviewees from the prepared adopters, Homeyer et al. (2021) show that suitable documentation of AI applications is essential for bringing AI out of medical laboratories and into hospitals. Rivera et al. (2020) and Liu et al. (2020) argue that the documentation of AI applications in medical research is essential and provide extensions to existing documentation guidelines to account for the unique nature of AI.

The fourth group of people, the unexcited non-adopters, do not use AI, even though they are working in use cases with low degrees of risk. However, due to the lack of interviewees in this group, the unexcited non-adopters could not be characterized in more detail. We invite researchers to investigate this group since it could give valuable insights into additional reasons for not adopting AI in practice.

The documentation approach presented by this group also shows that the documentation of AI can be an effective tool for ensuring accountability in the context of AI. The approach to documenting AI covers several challenges related to ensuring accountability presented by Diakopoulos (2016). First, the documentation enables auditors and regulators to inspect the human influence on and the human oversight over AI. Given the extensive documentation of the (potential) business implications of the integration of AI into the business process, it also becomes possible to evaluate the potential impact of specific human choices. Second, the detailed documentation of the training data, the data collection, and the data preparation process give auditors and regulators a clear picture of the quality of the data. Additionally, data subjects can dispute and correct incorrect observations about themselves. Third, the documentation is a suitable tool to communicate the presence of AI to auditors, regulators, and users. Knowledge about the presence of AI in business processes enables auditors, regulators, and users to judge if and how their lives could be influenced by AI and how they can fight against abuse. The approach for documenting AI is also suitable for ensuring ethical compliance in general in the context of AI. According to the interviewees' statements, documentation is an effective tool for ensuring the ethical implementation of AI.

From the perspective of documentation requirements, all groups agree that AI needs to be documented differently than traditional software. Particularly the inclusion of and focus on information about the training data and model development of the AI application makes the documentation of AI applications different from the documentation of traditional software and IT. Interviewees who are not using AI due to the high level of criticality of their use cases acknowledge that suitable AI documentation is a legal requirement. This group describes their requirements very much from an auditor's perspective and highlights that AI's actions and outcomes must be reproducible and transparent. The interviewees who use AI acknowledge the limitations of their approaches in regards to accountability and name key requirements.

This paper does not come without limitations. First, the interview partners for this study came from different industries. Given the background of the interview partners, this study provides a good overview of how AI is documented in use cases with high levels of criticality but may suffer from some degree of bias towards the banking industry. Second, there are no non-users of AI who work in use cases with a low degree of criticality in our study. Due to the scope of the study, this group of people was not targeted. However, this does not mean that this group does not exist. In a different research context, this group of people may be able to provide valuable insights to researchers. Third, our study had an exploratory nature, and thus we can not claim generalizability. However, our interview results saturated along the three investigated groups, and those groups could also be related to literature.

6. Conclusion

This paper provides a more nuanced understanding of why and in which type of use cases AI is currently documented and adopted. We identified missing AI-specific documentation and auditing guidelines as a significant barrier to adoption in use cases facing particular regulatory requirements or business risks. The contribution is valuable since they provide researchers and practitioners with a clear view of how AI is currently documented in practice and what are the adoption barriers. Our results also highlight the weaknesses and challenges of current AI documentation approaches, which are a prerequisite for AI adoption in use cases with high levels of criticality and high compliance requirements.

The insights from this interview study point to several interesting avenues for future research. Research on meaningful documentation guidelines for auditing AI seems promising. Interdisciplinary research considering the IS and the audit perspective seems promising from a managerial and a scientific point of view. Particularly, research on how applications of AI in regulated use cases can be audited suitably is scarce. Our results show that auditing is a prerequisite for adoption in use cases with high compliance requirements. This also calls for more nuanced research on AI adoption, taking documentation and compliance requirements into account.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Appendix A – Interview guideline

Interview Foundation

The Research Project: The project "Documenting Artificial Intelligence (AI) in the context of Commercial Banks" is the dissertation project of Florian Königstorfer at the University of Graz. The project aims to develop guidelines for documenting AI that support fairness, accountability, and transparency. The focus will be on applications of AI in commercial banks. Research has shown that AI can be applied in all of the core business areas of commercial banks and that the issue of explaining and documenting AI is a significant barrier for implementing AI in commercial banks.

Situation: Fairness, accountability, and transparency are not only essential topics in academic literature but also a necessary part of legislation relating to the processing of personal data. In the area of software development, proper documentation has been shown to enable the required degree of fairness, accountability, and transparency. Yet, unlike in the field of software development, no research could be found on how to design documentation standards that ensure that fairness, accountability, and transparency in the context of AI.

Study Goals: In this study, we aim to get an understanding of the requirements for AI documentation from a practitioner's point of view. To that end, interviews will be conducted with practitioners from commercial banks.

This study is the second in a series of six research papers including 1) a literature review on the use of AI in commercial banks; 2) an interview study on requirements for AI documentation in commercial banks; 3) a literature review on ways to achieve the requirements for AI documentation; 4) a case study on requirements for the documentation framework; 5) the proposal of a documentation framework, and 6) the evaluation of the documentation framework.

Interviewers: We are researchers at the Busines Analytics and Data Science Center at the University of Graz.

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Interview Guideline

1) Background of the Study

In recent years, the challenge of ensuring fairness, accountability, and transparency in the context of AI has become more and more pressing in academia as well as in practice. One way to work towards fair, accountable, and transparent AI, is to document the AI. However, little is known about how AI needs to be documented. This study intends to fix this issue by investigating the requirements for good AI documentation, and the limitations & challenges of creating good AI documentation.

1) Context and Scope

➢ Please describe your role in your organization and your educational background.

➢ What does the term AI mean to you? Please elaborate on your knowledge about and exposure to AI.

➢ Please describe 2-3 (existing or planned) use cases of AI in your organization.

➢ Please describe how AI is currently documented or how you plan to document AI.

1) Requirements for AI Documentations

➢ How satisfied are you with the current method for documenting AI?

➢ In your professional opinion, what are the requirements for good AI documentations?

Please elaborate on why you consider those aspects requirements.

1) Challenges and Limitations of AI Documentations

➢ For each use case, please describe the current challenges that you face when it comes to documenting AI and elaborate on why you consider those aspects challenging.

➢ Please name three limitations of current methods for documenting AI and elaborate on why you consider those aspects limitations.

1) General Outlook and Trends

➢ Please tell us a little bit about two of the most essential/disruptive current trends when it comes to documenting AI.

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